Clean Data
Better Analysis
Brighter Insights

ATGCATGCCTGA
GCTAGTCAGTTA
TTGACCTGAATC
CGTACGTTTAGCC
AGTCCGTAATGC
...

Data Preprocessing & Trimming Concepts

From Raw Sequencing Reads to High-Quality Data

RNA-seq Workshop | Bioinformatics & Computational Biology

Sequencing
Generate raw reads

```
@SEQ_ID
ACGTTGCAAGTCCGAT...
+
IIIIIIIIHHHHHHFFF...
```

Raw FASTQ Reads
Contains adapters, low-quality bases, and noise

Trimming & Filtering
Remove adapters, low-quality reads and contaminants

```
@SEQ_ID
ACGTTGCAAGTCCGAT...
+
IIIIIIIIIIIII...
```

High-Quality Reads
Ready for downstream analysis (alignment, quantification, etc.)

Adapter Removal | Quality Filtering | Remove Contaminants | Clean Reads | Reliable Results

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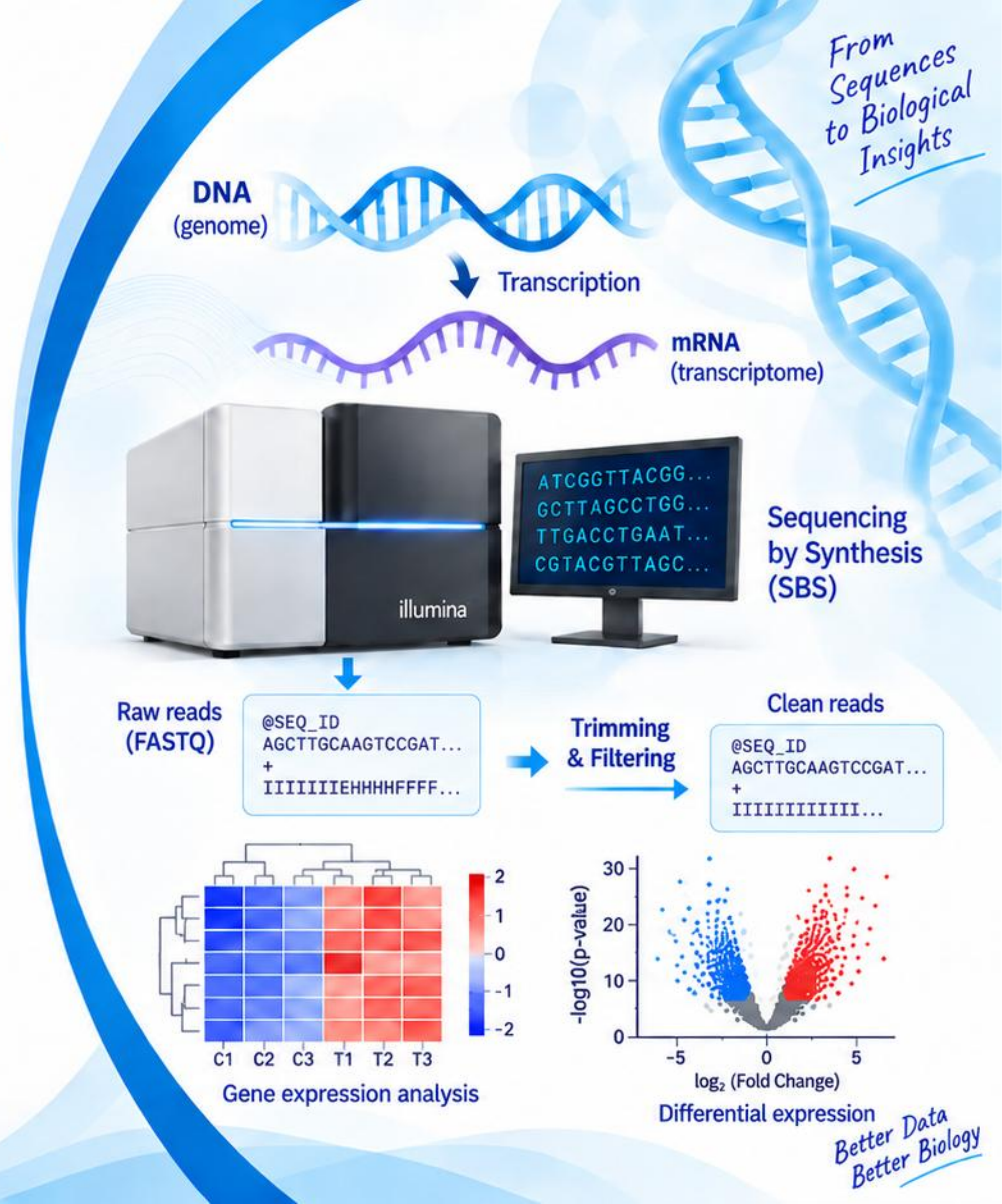
Empowering Through ICT

Quality Data
Drives Discovery

LEARNING OBJECTIVES

By the end of this session, you will be able to:

- Evaluate sequencing data quality using **FastQC**, and **MultiQC**, and interpret key quality metrics (e.g. per-base quality, GC content, duplication, adapter content).
- Perform **adapter** removal and quality trimming using **Trimmomatic** and **Cutadapt**, and assess the improvement in data quality after preprocessing.
- Generate and interpret **summary statistics** such as read counts, base composition and sequence length distribution to assess data suitability for downstream analysis.
- Understand how preprocessing and data quality **influence downstream RNA-seq analysis**, including read alignment, transcript quantification and differential expression.



RNA-seq Data → Information → Discovery

RNA-seq File Ecosystem

From Sequencer to Biological Insight

Different Files Connected for a Bigger Story!

1 Complete RNA-seq Workflow and File Flow

Sequencing & Data generation

ILLUMINA SEQUENCER (e.g., NovaSeq, NextSeq) → BCL (raw output)

Demultiplexing (bcl2fastq / Illumina DRAGEN) → FASTQ / FASTQ.GZ (sequence + quality)

Preprocessing & Quality control

Quality control (FASTQC)

Read trimming (Trimmomatic, fastp)

CLEAN FASTQ (trimmed reads)

Reference & Annotation

>chr1 ATGCGT... >chr2 TTGACA...

Reference genome (FASTA) (e.g., GRCh38.fa)

GTF

Gene annotation (GTF/GFF3) (e.g., genes.gtf)

Alignment & Processing

Alignment (STAR / HISAT2)

SAM / BAM (aligned reads) → BAM index (sample.bam.bai)

Read counting (featureCounts)

Quantification & Analysis

COUNT MATRIX (genes × samples)

Differential expression (DESeq2 / edgeR / limma-voom)

DEG TABLE (differentially expressed genes)

Biological Interpretation

Visualization (heatmap, volcano plot)

Functional analysis (GO, KEGG, pathways)

Biological insights (e.g., biomarkers, pathways, mechanisms)

2 Key RNA-seq Files and Their Meaning

File type	Meaning / What it contains	Main use in RNA-seq	Example
.bcl	Base call (raw signal from sequencer)	Raw Illumina output	Sample.bcl
.fastq	Sequence + quality (4-line format)	Primary RNA-seq reads	sample_R1.fastq
.fastq.gz	Compressed FASTQ	Storage / analysis	sample_R1.fastq.gz
.html / .zip	QC report (FASTQC)	Quality assessment	fastqc_sample.html
.fa / .fasta	Reference genome (FASTA format)	Alignment	GRCh38.fa
.gtf	Gene annotation (genes, transcripts, exons)	Gene-level quantification	genes.gtf
.gff3	Genome annotation (genome features)	Genome feature information	genes.gff3
.sam	Text alignment file	Alignment output (human readable)	sample.sam
.bam	Binary alignment file (compressed)	Downstream analysis	sample.bam
.bai	BAM index	Rapid genomic access (index for BAM)	sample.bam.bai
.txt / .tsv	Count table (genes × samples)	Gene quantification	counts.txt
.csv / .tsv	DEG results	Statistical results	DEG_results.csv
.png / .pdf / .svg	Figures (plots)	Visualization (heatmap, volcano, etc.)	heatmap.png

3 Example Files at Different Stages

Raw sequencing output (BCL) — BCL: Sample.bcl

Raw reads (FASTQ) — FASTQ: Sample_R1.fastq.gz, Sample_R2.fastq.gz

QC report — HTML: fastqc_sample.html

Reference & annotation — FASTA: GRCh38.fa, genes.gtf

Alignment output — BAM: sample.bam, sample.bam.bai

Count matrix — TXT: counts.txt

Differential expression — CSV: DEG_results.csv

Visualization — heatmap.png, volcano.png

4 Key Take-Home Messages

1. **FASTQ** tells us what was sequenced.
2. **BAM** tells us where the reads aligned.
3. **GTF** tells us what genomic features are present.
4. **Count matrix** tells us how many reads belong to each gene.
5. **DEG table** tells us which genes changed between conditions.

5 Common Tools Used

Step	Example tools
Demultiplexing	bcl2fastq, Illumina DRAGEN
Quality control	FastQC, MultiQC
Trimming	Trimmomatic, fastp
Alignment	STAR, HISAT2
Read counting	featureCounts, HTSeq-count
Differential expression	DESeq2, edgeR, limma-voom
Functional analysis	clusterProfiler, g:Profiler, DAVID

Raw Data (BCL) → Sequences (FASTQ) → Quality Control (HTML) → Reference (FASTA / GTF) → Alignment (SAM/BAM) → Counts (TSV) → Differential Expression (CSV) → Biological Insights (Plots & Pathways)

Files to Find Answers!

Downloading SRA Data & FASTQ Files

Two common routes for RNA-seq data download

Same Data
Different Routes
Same Goal!

A Download using SRA Toolkit (NCBI)

SRA Toolkit is used to download SRA files (SRR) and convert them to FASTQ format.

1 Download SRA file

```
prefetch SRR4244296
```

This downloads the .sra file to your local machine.

2 Convert SRA → FASTQ

```
fasterq-dump SRR4244296 --split-files
```

Converts .sra file to FASTQ. --split-files gives separate files for paired-end reads.

3 Compress FASTQ files (optional)

```
gzip SRR4244296_*.fastq
```

Compress to save disk space.

4 Paired-end output

```
SRR4244296_1.fastq.gz
SRR4244296_2.fastq.gz
```

Output files for paired-end sequencing.

5 Use multiple threads (for large datasets)

```
fasterq-dump SRR4244296 --split-files -e 8
```

-e 8 → uses 8 threads (faster).

Tip:

- Install SRA Toolkit from NCBI (https://github.com/ncbi/sra-tools)
- Check available space before downloading large datasets.

Workflow: Two Routes to FASTQ

SRA / ENA ACCESSION
(e.g., SRR4244296)

SRA TOOLKIT (NCBI)	ENA (EMBL-EBI)
NCBI	ENA European Nucleotide Archive
SRR number (e.g., SRR4244296)	**Direct FASTQ files** (from ENA FTP/HTTP)
prefetch + fasterq-dump	wget / curl
FASTQ files (SRR4244296_1.fastq.gz) (SRR4244296_2.fastq.gz)	**FASTQ files** (sample_R1.fastq.gz) (sample_R2.fastq.gz)

FASTQ DATA

Quality control (FastQC)

B Direct FASTQ Download from ENA (EMBL-EBI)

ENA provides FASTQ files directly, so you can skip the SRA → FASTQ conversion step.

1 Search in ENA Browser

- Go to https://www.ebi.ac.uk/ena/browser/
- Search using the SRR accession (e.g., SRR4244296)
- Open the run page and look for FASTQ files.

ENA European Nucleotide Archive — SRR4244296

2 Get FASTQ download link

- Select the FASTQ files (usually _1.fastq.gz and _2.fastq.gz).
- Copy the FTP or HTTP link.
- Links look like:

```
https://ftp.sra.ebi.ac.uk/vol1/fastq/SRR424/006/SRR4244296/
SRR4244296_1.fastq.gz
```

3 Download using wget or curl

```
wget "https://ftp.sra.ebi.ac.uk/vol1/fastq/SRR424/006/SRR4244296/
    SRR4244296_1.fastq.gz"
wget "https://ftp.sra.ebi.ac.uk/vol1/fastq/SRR424/006/SRR4244296/
    SRR4244296_2.fastq.gz"
# or using curl
curl -O "https://ftp.sra.ebi.ac.uk/vol1/fastq/SRR424/006/SRR4244296/
    SRR4244296_1.fastq.gz"
```

Advantages of ENA direct download

- ✓ No SRA → FASTQ conversion required
- ✓ Usually faster and simpler
- ✓ Convenient for large datasets
- ✓ Provides compressed FASTQ files (.fastq.gz)

Comparison of the Two Methods

Feature	SRA Toolkit (NCBI)	ENA Direct Download (EMBL-EBI)
Input	SRR accession	SRR accession
What you download	SRA file (.sra)	FASTQ files (.fastq.gz)
Conversion step	Required (prefetch + fasterq-dump)	Not required
Command	prefetch + fasterq-dump	wget / curl
Speed	Slower (conversion step)	Usually faster
Use case	Useful for SRA archives	Convenient for direct FASTQ retrieval

Next Steps After Download

1. FASTQ data (.fastq.gz)
2. Quality control (FastQC)
3. Trimming (e.g., Trimmomatic)
4. QC again (FastQC)
5. Alignment (e.g., HISAT2)

Key Take-Home Message

Both methods give you FASTQ files. Choose the one that is convenient for your data and analysis.

From Data to Discovery!

FastQC: Quality Control of RNA-seq Data

Assess the quality of your sequencing reads before downstream analysis

Good Data Better Results!

1 What is FastQC?

FastQC is a Java-based quality-control tool that performs a series of statistical analyses on raw sequencing reads in FASTQ/FASTQ.GZ format.

- Open-source tool (Babraham Institute)
- Works on FASTQ / FASTQ.GZ files
- Provides an easy-to-interpret HTML report
- Widely used for RNA-seq, genomic and metagenomic data

2 Why FastQC is Used

- Assess sequencing/base-call quality
- Detect low-quality bases and reads
- Identify adapter contamination
- Examine GC-content and nucleotide composition
- Detect sequence duplication
- Identify overrepresented sequences and k-mers
- Compare data quality before and after trimming

3 FastQC Input → Output

RAW FASTQ / FASTQ.GZ

sample_R1.fastq.gz
sample_R2.fastq.gz
...
*.fastq.gz

→ **FastQC** (Java-based QC tool) →

- Quality statistics
- Sequence statistics
- Adapter analysis
- GC / nucleotide analysis
- Duplication / k-mer analysis

HTML Report + ZIP File

sample_fastqc.html
sample_fastqc.zip

4 Basic Linux Command

Run FastQC on a single file:

```
fastqc sample_R1.fastq.gz -o fastqc_results
```

Run FastQC on multiple files (using 8 threads):

```
fastqc *.fastq.gz -t 8 -o fastqc_results
```

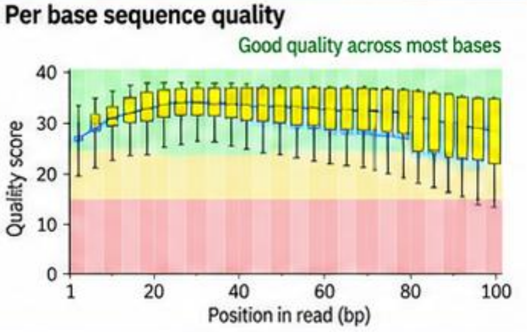
- `-t 8` = use 8 processing threads
- `-o` = specify output directory

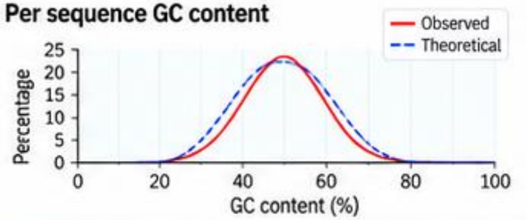
```
user@linux:~$ fastqc *.fastq.gz -t 8 -o fastqc_results
Started analysis of sample_R1.fastq.gz
Analysis complete for sample_R1.fastq.gz
Started analysis of sample_R2.fastq.gz
Analysis complete for sample_R2.fastq.gz
```

5 Example FastQC Report (Key Modules)

FastQC Report

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content





6 What FastQC Reports

Module	What it Tells You
Basic Statistics	Summary of read counts, length, encoding etc.
Per base sequence quality	Quality score at each base position
Per tile sequence quality	Detects spatial bias on flowcell
Per sequence quality scores	Distribution of average quality per read
Per base sequence content	A, T, G, C content at each position
Per sequence GC content	GC content distribution across reads
Per base N content	Proportion of ambiguous bases (N)
Sequence length distribution	Distribution of read lengths
Sequence duplication levels	Level of duplicated reads
Overrepresented sequences	Identifies overrepresented reads/k-mers
Adapter content	Detects presence of adapter sequences

Important: FastQC does not trim, filter, or alter FASTQ reads. It only analyzes and reports their characteristics.

Check Quality Before You Proceed!

FastQC: Backend Algorithm & Statistical Processing

FastQC — A quality control tool for high-throughput sequencing

Inside the black box: How FastQC analyzes your sequencing reads

From FASTQ to Quality Insights!

1 FASTQ Parsing

FastQC reads each FASTQ record (4 lines):

```
@SRR4244296.1      →  Read identifier
ACGTACGTACGT       →  DNA sequence
+                  →  Separator
IIIIIIIIIIII       →  Quality string
```

- The sequence and quality strings are extracted separately.
- Supports FASTQ and compressed FASTQ.GZ files.

2 Quality Score Conversion (Phred)

FastQC converts ASCII characters to **Phred quality** scores.

For Illumina (Phred+33):

$$Q = \text{ASCII value} - 33$$

Error probability:

$$P_e = 10^{-Q/10}$$

Example:

$$Q30 \rightarrow P_e = 10^{-30/10} = 0.001 = 0.1\%$$

- Higher Q = better base call quality.
- FastQC automatically detects the encoding (Phred+33 or Phred+64).

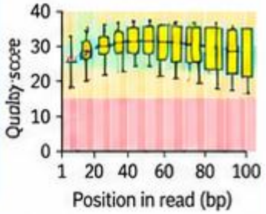
3 Per-base Quality Calculation

For each position i, FastQC calculates quality statistics across all reads:

$$\bar{Q}_i = \frac{1}{N}\sum_{j=1}^{N} Q_{ij}$$

Statistics calculated:

- Mean
- Median
- Quartiles (25% and 75%)
- 10th and 90th percentiles



This generates the **Per Base Sequence Quality plot.**

4 Overall Workflow of FastQC (Internal Processing)

FASTQ / FASTQ.GZ
sample_R1.fastq.gz
sample_R2.fastq.gz

↓

File Decompression
(if .gz)

↓

FASTQ Parser
Read 4-line records

↓

DNA Sequence
ACGTACGTACGT
(Nucleotides)

Quality String
IIIIIIIIIIII
(ASCII → Phred Q)

↓

Statistical Modules
Calculate various quality and composition metrics

↓

Per-base Quality | **GC / Base Composition** | **Sequence Duplication / K-mers** | **Adapter Analysis**

↓

Threshold / Distribution Analysis
Compare against built-in QC rules

↓

PASS / WARN / FAIL

↓

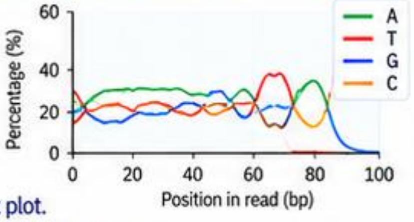
HTML REPORT
Interactive plots and summary

5 Sequence Composition Analysis

At each position i, FastQC calculates the proportion of each nucleotide:

$$\%A_i = \frac{N_A(i)}{N(i)} \times 100$$

Similarly for T, G and C.



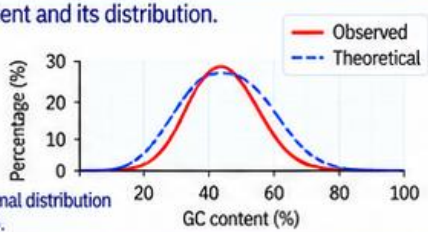
- Produces Per Base Sequence Content plot.
- Deviations from expected composition may indicate adapter contamination, biased libraries or other issues.

6 GC Content Analysis

FastQC calculates the overall GC content and its distribution.

$$GC\% = \frac{(N_G + N_C)}{N} \times 100$$

- Produces Per Sequence GC Content plot.
- Compares observed distribution with a normal distribution (expected for random genomic sequences).



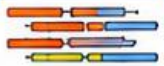
7 Adapter and Overrepresented Sequences

- FastQC searches for known adapter sequences using exact or partial matching (k-mer based).
- Counts the occurrence of overrepresented sequences and k-mers.
- Reports their abundance and position in reads.

Adapter sequence detected

8 Duplication Analysis

- Counts the number of identical reads.
- Calculates the percentage of duplicate sequences.
- High duplication may indicate PCR bias or low library complexity.



Duplicate reads

9 Key Take-Home Message

- FastQC is a statistical analysis and QC pipeline.
- It does not trim, filter, or alter FASTQ reads.
- It helps you identify potential issues before downstream analysis.
- Always run FastQC before and after trimming.

Better QC
Better Data
Better Science!

FastQC: Interpreting All Parameters for RNA-seq Data

From raw reads to reliable downstream analysis

Check Quality
Understand Your Data
Make Better Decisions!

1 FastQC: Basic Statistics & Quality Scores

Read quality is the foundation of reliable analysis

1.1 Basic Statistics

Metric	Example value
Total reads	20,000,000
Read length	150 bp
Total bases	3,000,000,000
GC percentage	48.2 %
Quality encoding	Phred+33

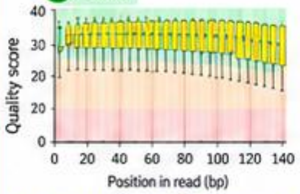
Good
- Expected read length
- Plausible GC%

Bad
- Unexpected read length
- Extremely unusual GC%
- Very few reads

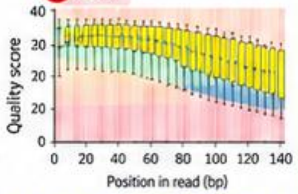
2 Per Base Sequence Quality

$$Q = -10\log_{10}(P_e) \qquad \bar{Q}_i = \frac{1}{N}\sum_{j=1}^{N} Q_{ij}$$

Good



Bad



- Quality remains high across read
- Q30 or higher is excellent
- Strong quality decline at 3′ end
- Large proportion of Q < 20

Possible cause: Sequencing-cycle quality deterioration.

3 Per Sequence Quality Scores

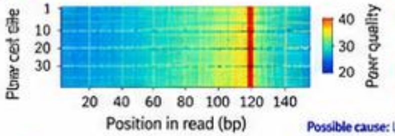
$$Q_{read} = \frac{1}{L}\sum_{i=1}^{L} Q_i$$

Good

Bad

- Most reads have high average quality
- Large peak of low-quality reads

4 Per Tile Sequence Quality



Good
- Similar quality across tiles

Bad
- One or more tiles with substantially lower quality

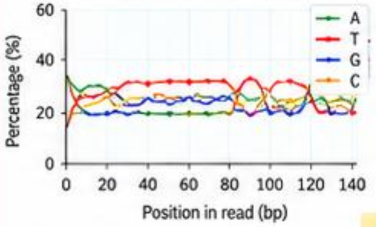
Possible cause: Localized sequencing/imaging problem.

2 FastQC: Sequence Composition & GC Content

Understand the composition of your reads

5 Per Base Sequence Content

$$\%A_i = \frac{A_i}{A_i + T_i + G_i + C_i} \times 100$$



Good
- Relatively stable composition

Bad
- Strong positional nucleotide bias

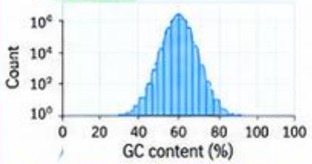
Possible causes:
- Random-priming bias
- Library preparation bias
- Contamination
- Biological sequence bias

RNA-seq may naturally show nucleotide bias.

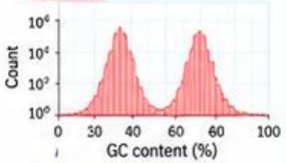
6 Per Sequence GC Content

$$GC\% = \frac{G + C)}{A + T + G + C} \times 100$$

Good



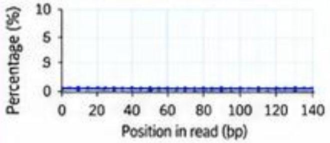
Bad



- Expected single smooth peak
- Multiple or unusual peaks

Possible causes: Contamination, mixed organsims, PCR bias, unusual library composition.

7 Per Base N Content

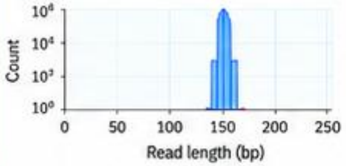


Good ~0% N

Bad Increasing/high N

Possible cause: Poor base-calling or sequencing signal.

8 Sequence Length Distribution



Good
Expected uniform length (e.g. ~150 bp)

Bad
Unexpected short/ variable lengths

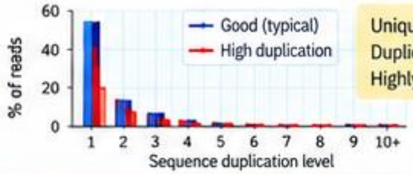
Possible causes: Trimming, fragmentation, or library problems.

3 FastQC: Duplication, Overrepresentation & K-mers

Detect redundant and unusual sequences

9 Sequence Duplication Levels

Counts identical sequences and estimates duplication.



Unique read → 1×
Duplicated read → 2×
Highly duplicated → 100×

Good
- Low/moderate duplication

Bad
- Very high duplication

Possible causes:
- PCR amplification
- Low-complexity library
- Highly abundant transcripts

RNA-seq: High duplication does not automatically mean poor data (highly expressed transcripts).

10 Overrepresented Sequences

Identifies sequences occurring at unusually high frequency.

Sequence	Count	% of reads	Possible source
AGATCGGAAGAGC	125,432	0.63%	Adapter
AAGTTT...	98,234	0.49%	rRNA/contamination
GACTAC...	76,543	0.38%	Highly expressed
...	...	...	...

Good
- Few/no unexpected sequences

Bad
- Large number of specific sequences

Possible sources: Adapter sequences, rRNA, highly expressed transcripts, contamination.

11 K-mer Content

Counts k-mers and detects unusually enriched sequences.

Example 5-mers:
ACGTA
CGTAC
GTACG
TACGT

K-mer	Count	Enrichment
GATCG	152,324	12.5
ATCGG	143,221	11.8
TCGGA	131,004	10.2
CGGAA	120,987	9.7

Good
- No strong enrichment

Bad
- Strong enrichment of specific k-mers

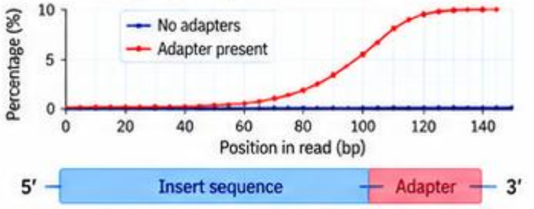
Possible causes: Adapter/primer contamination, PCR bias, low-complexity sequences, highly abundant transcripts.

4 FastQC: Adapter Content, Status & Interpretation

Detect residual adapters and summarize overall quality

12 Adapter Content

Searches for known adapter sequences and calculates their prevalence.



Good
Adapter content ≈ 0%

Bad
Adapter content increases toward 3′ end

Possible cause: Insert is shorter than sequencing read.
Action: Trim using Cutadapt, Trimmomatic or fastp.

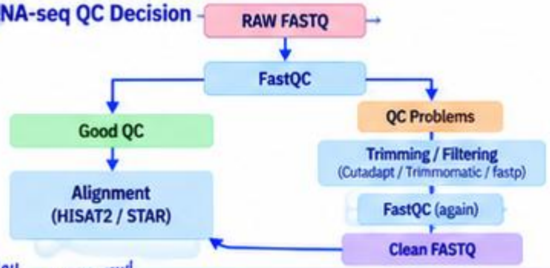
PASS / WARN / FAIL

Status	Meaning	Typical interpretation
PASS	Within expected limits	Good
WARN	Potential deviation	Investigate
FAIL	Strong deviation	Investigate/preprocess

Good vs Bad FastQC Profile

Parameter	Good	Problematic
Per-base quality	High, stable	Strong 3′ decline
Mean read quality	High	Large low-Q population
GC distribution	Expected/smooth	Multiple unusual peaks
N content	~0%	High N
Read length	Expected	Unexpected variable lengths
Duplication	Low/moderate	Very high
Adapter	Near 0%	High/increasing
Overrepresented sequences	Few/none	Many
K-mers	No strong enrichment	Strong enrichment

16 RNA-seq QC Decision



Key message: FastQC detects and summarizes quality problems; it does not correct them.

Quality Today
Better Biology Tomorrow!

FastQC
A quality control tool for high-throughput sequencing

FastQC: Essential Documentation & Resources

Official resources, technical details and further reading for effective use of FastQC

Better Data
Better Science!

1 Official FastQC Documentation

FastQC Official Documentation

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

- Installation, operation and general FastQC information
- Overview of features and requirements
- Useful for getting started

2 FastQC Analysis Modules

Analysis Modules Documentation

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/Help/3 Analysis Modules/

- Detailed explanation of all analysis modules
- Per Base Quality, GC Content, Duplication, Adapter Content, K-mers, etc.
- Helps understand what each plot means and how it is generated

3 Evaluating Results (PASS / WARN / FAIL)

PASS / WARN / FAIL Interpretation

https://www.bioinformatics.babraham.ac.uk/projects/fastqc/Help/2 Basic Operations/2.2 Evaluating Results.html

- Explains how FastQC determines PASS, WARN and FAIL status
- Guidelines for interpreting QC results
- Helps in decision making for further analysis (e.g., trimming)

4 FastQC Source Code (GitHub)

FastQC GitHub Repository

https://github.com/s-andrews/FastQC

- Java source code of FastQC
- Backend architecture and analysis modules
- Configuration files and report generation
- Useful for understanding how FastQC works internally

5 Benchmarking Paper

de Sena Brandine & Smith (2021)

Falco: high-speed FastQC emulation for quality control of sequencing data.

Bioinformatics, 37(19), 3222–3224 (2021).

doi: 10.1093/bioinformatics/btab446

- Direct benchmarking of FastQC using real sequencing datasets
- Provides insights into FastQC analyses and computational performance
- Useful for understanding FastQC algorithms and limitations

Learning Path (Recommended Order)

Official Documentation → Analysis Modules → PASS / WARN / FAIL Interpretation → Source Code (GitHub) → Benchmarking Paper

Recommended for This Workshop:

Start with the Official Documentation, explore the Analysis Modules, understand PASS/WARN/FAIL interpretation, look at the source code for deeper insight, and finally read the benchmarking paper for advanced understanding.

"Good quality data leads to reliable analysis."

FastQC Alternatives: fastp, Falco & MultiQC

QC Beyond FastQC

Faster, more features and multi-sample reporting for sequencing data quality control

Multiple Tools Better Insights!

1 fastp — QC + Trimming + Filtering

All-in-one solution for preprocessing and quality control

Fast · Comprehensive · Popular

fastp

- Performs quality control, adapter trimming and filtering
- Supports single-end and paired-end RNA-seq
- Generates interactive HTML/JSON reports
- Very fast and memory efficient
- Widely used for large-scale datasets

APT installation (Ubuntu/Debian)

```
sudo apt update
sudo apt install fastp
```

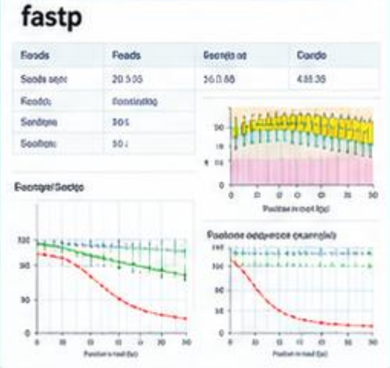
Conda installation

```
conda install -c bioconda fastp
```

Example command (paired-end):

```
fastp -i sample_R1.fastq.gz \
    -I sample_R2.fastq.gz \
    -o clean_R1.fastq.gz \
    -O clean_R2.fastq.gz \
    -h fastp.html \
    -j fastp.json
```

fastp HTML Report (example)



Output files:

- Clean FASTQ files
- HTML report (fastp.html)
- JSON report (fastp.json)

2 Falco — FastQC-Compatible QC

High-speed alternative to FastQC

Fast · Efficient · FastQC-compatible

Falco – Speed up your QC –

- High-performance alternative to FastQC
- Produces FastQC-compatible results
- Designed for large and high-throughput datasets
- Much faster than FastQC
- Easy to use and lightweight

APT installation (Ubuntu/Debian)

```
sudo apt update
sudo apt install falco
```

Conda installation

```
conda install -c bioconda falco
```

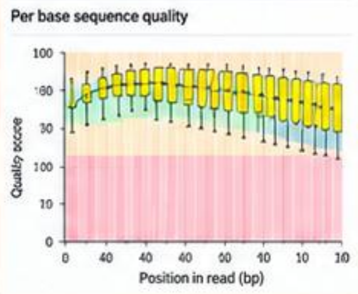
Example command:

```
falco sample.fastq.gz
```

Output files:

- HTML report (FastQC-compatible)
- ZIP file with detailed results
- Same modules and plots as FastQC

Falco report (example)



3 MultiQC — Multi-Sample QC Report

Aggregate results from multiple tools and samples

Integrate · Summarize · Visualize

MultiQC

- Collects results from multiple bioinformatics tools
- Supports FastQC, fastp, HISAT2, STAR, etc.
- Combines results from many samples into one report
- Generates a single interactive HTML report
- Ideal for large projects and workshops

APT / pip installation (Ubuntu/Debian)

```
sudo apt update
sudo apt install python3-pip
pip3 install multiqc
```

Conda installation

```
conda install -c bioconda multiqc
```

Example command:

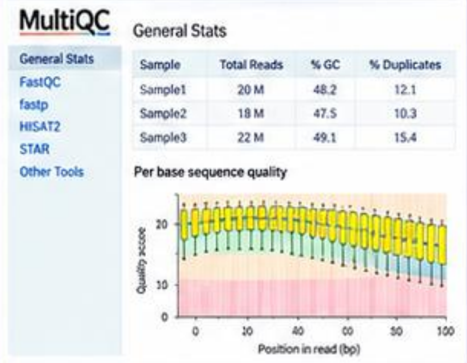
```
multiqc .
```

\# Run in the directory containing all QC results

Output files:

- multiqc_report.html
- multiqc_data/ (raw data)
- Summary of all samples
- Interactive plots and tables

MultiQC report (example)



About Conda

CONDA

- Conda is an open-source package and environment manager (Anaconda/Miniconda).
- Helps install and manage bioinformatics tools and their dependencies.
- Creates isolated environments to avoid software conflicts.
- **Bioconda** is a community channel with many bioinformatics tools.

Useful Conda commands:

```
conda create -n rnaseq                 # create a new environment
conda activate rnaseq                  # activate environment
conda install -c bioconda <tool>       # install a tool
conda deactivate                       # deactivate environment
```

Different Tools Same Goal — Better Data, Better Science!

Errors Happen
Solutions Exist!

Basic Errors You May Encounter During RNA-seq QC

Common problems, causes and solutions for FastQC, fastp, Falco and MultiQC

Troubleshoot Today
Run Smoothly
Tomorrow!

1 Software Not Installed

```
fastqc: command not found
fastp: command not found
falco: command not found
multiqc: command not found
```

Cause: Software is not installed or not in PATH.

Solution:
- Install the required software (apt, conda or pip).
- Check installation with:

```
fastqc --version
```

2 FASTQ File Not Found

```
ERROR: Can't open file
Cancer1_R1.fastq.gz
```

Cause: Wrong directory or incorrect filename.

Solution:
- Check current directory:

```
pwd
```

- List files:

```
ls *.fastq.gz
```

3 Output Directory Missing

```
No such file or directory
```

Cause: Output directory does not exist.

Solution:
- Create the required directories:

```
mkdir -p QC/raw_fastqc \
QC/raw_falco trimmed
```

4 Corrupted FASTQ File

```
gzip: unexpected end of file
```

Cause: Incomplete or corrupted compressed file.

Solution:
- Check file integrity:

```
gzip -t Cancer1_R1.fastq.gz
```

- Re-download the file if corrupted.

5 Insufficient Disk Space

```
No space left on device
```

Cause: Not enough storage space for processing or output files.

Solution:
- Check available disk space:

```
df -h
```

- Remove unnecessary files or **use a different storage location.**

6 R1/R2 Pairing Error

```
Correct:
 Cancer1_R1.fastq.gz
 Cancer1_R2.fastq.gz
Incorrect pairing:
 Cancer1_R1.fastq.gz
 Cancer2_R2.fastq.gz
```

Cause: R1 and R2 files from different samples are used.

Solution:
- Always keep R1 and R2 from the same sample together.

7 Permission Denied

```
Permission denied
```

Cause: No permission to read input files or write output files.

Solution:
- Check file permissions:

```
ls -l
```

- Change permissions if needed:

```
chmod +r file.fastq.gz
```

8 MultiQC Finds No Results

```
No analysis results found
```

Cause: QC output directories are empty or incorrect paths were provided.

Solution:
- Check that QC results exist:

```
ls QC/raw_fastqc
ls QC/trimmed_fastqc
```

- Run MultiQC in the correct directory.

9 Java Error with FastQC

```
Unable to access jarfile
fastqc.jar
or other Java-related errors.
```

Cause: Java is not installed or incorrectly configured.

Solution:
- Check Java installation:

```
java -version
```

- Install Java if required (e.g. openjdk).

10 WSL Path Error

```
Windows path:
 E:\RNAseq
WSL path:
 /mnt/e/RNAseq
```

Cause: Using Windows path format inside WSL.

Solution:
- Use the correct WSL path.
- Check current location:

```
pwd
ls
```

Key Troubleshooting Checklist

Before running RNA-seq QC, verify:

- Software installed
- Sufficient disk space
- Correct working directory
- Output directories exist
- FASTQ filenames are correct
- FASTQ files are not corrupted
- R1/R2 pairs match

Check early
Save time
Avoid frustration!

AGATCGGAAGAGC
AGATCGGAAGAGC
AGATCGGAAGAGC

Trimmomatic: Installation & Read Trimming

Remove adapters and low-quality bases for high-quality RNA-seq analysis

Clean Reads Better Results!

1 Installation in Ubuntu / WSL

Install Trimmomatic using sudo:

```
$ sudo apt update
$ sudo apt install trimmomatic
```

Check installation:

```
$ trimmomatic -version
```

✅ If installed correctly, you will see the Trimmomatic version information.

2 Important Trimming Parameters

Parameter	Function	Setting / Example
PE	Paired-end mode	Paired reads
-threads 4	CPU threads	4
ILLUMINACLIP	Removes adapter sequences	TruSeq3-PE
2:30:10	Adapter clipping thresholds	Seed: 2 Palindrome: 30 Simple: 10
LEADING:3	Removes low-quality bases at 5′ end	Q < 3
TRAILING:3	Removes low-quality bases at 3′ end	Q < 3
SLIDINGWINDOW:4:20	Quality-based trimming	4 bp / Q20
MINLEN:36	Removes very short reads	< 36 bp

3 Example Command (Paired-end)

```
trimmomatic PE -threads 4 \\
  sample_R1.fastq.gz sample_R2.fastq.gz \
  trimmed/sample_R1_paired.fastq.gz \
  trimmed/sample_R1_unpaired.fastq.gz \
  trimmed/sample_R2_paired.fastq.gz \
  trimmed/sample_R2_unpaired.fastq.gz \
  ILLUMINACLIP:TruSeq3-PE.fa:2:30:10 \
  LEADING:3 TRAILING:3 \
  SLIDINGWINDOW:4:20 \
  MINLEN:36
```

Note:

- Use the appropriate adapter file (e.g. TruSeq3-PE.fa).
- Adjust parameters based on your dataset and sequencing platform.
- Create the output directory before running the command:

```
mkdir -p trimmed
```

4 Trimming Workflow

Raw FASTQ
↓
Adapter Removal (ILLUMINACLIP)
↓
Quality Trimming (LEADING, TRAILING, SLIDINGWINDOW)
↓
Length Filtering (MINLEN)
↓
Trimmed FASTQ (R1_paired + R2_paired)
↓
HISAT2 Alignment (Downstream Analysis)

5 Output Files

Output File	Description
sample_R1_paired.fastq.gz	Trimmed paired reads (R1) → used for downstream analysis
sample_R1_unpaired.fastq.gz	Trimmed unpaired reads (R1) → usually discarded
sample_R2_paired.fastq.gz	Trimmed paired reads (R2) → used for downstream analysis
sample_R2_unpaired.fastq.gz	Trimmed unpaired reads (R2) → usually discarded

✅ **Paired reads:** R1_paired + R2_paired
→ used for downstream paired-end alignment (e.g. HISAT2).

6 Tips

- ✅ Use the correct adapter file for your sequencing platform.
- ✅ Check read quality with FastQC before and after trimming.
- ✅ Adjust parameters if needed (e.g. for low-quality or short reads).
- ✅ Keep paired reads for downstream analysis.
- ✅ Remove or ignore unpaired reads unless required.
- ✅ Run MultiQC to compare QC before and after trimming.

Trim smart, analyze better!